THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 6 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Patterns in Mathematics** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Powers of 2: $2, 4, 8, 16, \ldots$ The 9th term is: 1

(a) 768 **(b)** 510 **(c)** 512 **(d)** 514

2. The nth term of the pattern $5, 14, 23, 32, \ldots$ is $a + (n-1)d$. The 14th term is: 1

(a) 113 **(b)** 131 **(c)** 123 **(d)** 122

3. In the pattern $10, 16, 22, 28, \ldots$ which term number equals 136? 1

(a) 22 **(b)** 23 **(c)** 24 **(d)** 21

4. What is the sum of the first 22 even numbers $2 + 4 + 6 + \ldots$? 1

(a) 506 **(b)** 528 **(c)** 484 **(d)** 507

5. Which triangular number comes just after 496? 1

(a) 559 **(b)** 528 **(c)** 527 **(d)** 529

6. A pattern of stacked sticks uses 3 sticks for 1 triangle and 2 more sticks for each extra triangle added in a row. Sticks for 25 triangles: 1

(a) 53 **(b)** 51 **(c)** 52 **(d)** 50

7. The sum of the first 39 natural numbers added to the sum of the first 40 natural numbers is: 1

(a) 1639 **(b)** 1601 **(c)** 1599 **(d)** 1600

8. Powers of 2: $2, 4, 8, 16, \ldots$ The 10th term is: 1

(a) 1022 **(b)** 1024 **(c)** 1026 **(d)** 1536

9. The nth term of the pattern $16, 27, 38, 49, \ldots$ is $a + (n-1)d$. The 9th term is: 1

(a) 105 **(b)** 93 **(c)** 104 **(d)** 115

10. In the pattern $8, 11, 14, 17, \ldots$ which term number equals 98? 1

(a) 32 **(b)** 31 **(c)** 33 **(d)** 30

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of the first 40 odd numbers and the first 40 even numbers. 2

12. Find the nth term formula of $6, 11, 16, 21, \ldots$ and the 25th term. 2

13. Is 82 a term of $9, 17, 25, 33, \ldots$? Give a reason. 2

14. Find the sum of the first 50 odd numbers and the first 50 even numbers. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. How many triangles' worth of sticks: a row of 24 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 59 sticks. 3

16. Find the sum of all odd numbers from 1 to 119. Hence find the sum of all even numbers from 2 to 120. 3

17. Write the 36th and 37th triangular numbers and check that they add to a square. 3

OR

How many triangles' worth of sticks: a row of 27 triangles needs 3 sticks for the first triangle and 2 more for each new one. Find the sticks needed and the number of triangles that can be made with 65 sticks. 3

18. Find the sum of all odd numbers from 1 to 149. Hence find the sum of all even numbers from 2 to 150. 3

OR

Write the 40th and 41st triangular numbers and check that they add to a square. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A pattern of squares: the first figure has 6 squares and each next figure has 3 more. Find the number of squares in the 14th figure, and the first figure that has more than 95 squares. 5

OR

Show that the sum of the cubes of the first 13 numbers is a perfect square, and find the sum of the first 13 numbers. 5

20. A pattern of squares: the first figure has 2 squares and each next figure has 8 more. Find the number of squares in the 8th figure, and the first figure that has more than 108 squares. 5

OR

Show that the sum of the cubes of the first 14 numbers is a perfect square, and find the sum of the first 14 numbers. 5

END OF QUESTION PAPER — Total: 40 Marks

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